



Section I: Scenario Demographics

Scenario Title:	CCB OD
Date of Development:	24/05/2015 (DD/MM/YYYY)
Target Learning Group:	☐ Juniors (PGY 1 - 2) ☒ Seniors (PGY ≥ 3) ☐ All Groups

Section II: Scenario Developers

Scenario Developer(s):	Kyla Caners
Affiliations/Institution(s):	McMaster University
Contact E-mail (optional):	kcaners@gmail.com

Section III: Curriculum Integration

Learning Goals & Objectives	
Educational Goal:	To expose learners to a multi-drug overdose and the management of a severe CCB OD.
CRM Objectives:	1) Maintain calm leadership throughout complex case. 2) Demonstrate situational awareness as patient status changes
Medical Objectives:	1) Recognize a possible CCB overdose 2) Administer appropriate early treatments for a CCB overdose 3) Recognize hypercapneic respiratory failure and manage appropriately. 4) Consider intralipid as treatment in an arrested CCB overdose

Case Summary: Brief Summary of Case Progression and Major Events

A 48-year-old female presents with a possible multi-drug overdose including glyburide, clonazepam and nifedipine. She will remain hypotensive throughout the case, despite treatment with calcium, high dose insulin, and other vasopressors. She will also have progressive respiratory depression and will eventually require intubation. She will then proceed to arrest. The team will be expected to give intralipid once the patient has arrested.

References

Marx, J. A., Hockberger, R. S., Walls, R. M., & Adams, J. (2013). *Rosen's emergency medicine: Concepts and clinical practice*. St. Louis: Mosby.



Section IV: Scenario Script

A. Scenario Cast & Realism			
Patient:	☒ Computerized Mannequin	Realism:	☒ Conceptual
	☐ Mannequin		☐ Physical
	☐ Standardized Patient		☐ Emotional/Experiential
	☐ Hybrid		☐ Other:
	☐ Task Trainer		☐ N/A
<i>Select most important dimension(s)</i>			
Confederates	Brief Description of Role		
Family member	Brings list of medications, describes patient's mood.		
B. Required Monitors			
☒ EKG Leads/Wires	☐ Temperature Probe	☐ Central Venous Line	
☒ NIBP Cuff	☐ Defibrillator Pads	☐ Capnography	
☒ Pulse Oximeter	☐ Arterial Line	☐ Other:	
C. Required Equipment			
☒ Gloves	☒ Nasal Prongs	☐ Scalpel	
☒ Stethoscope	☒ Venturi Mask	☐ Tube Thoracostomy Kit	
☐ Defibrillator	☒ Non-Rebreather Mask	☐ Cricothyroidotomy Kit	
☒ IV Bags/Lines	☒ Bag Valve Mask	☐ Thoracotomy Kit	
☒ IV Push Medications	☒ Laryngoscope	☐ Central Line Kit	
☐ PO Tabs	☐ Video Assisted Laryngoscope	☐ Arterial Line Kit	
☐ Blood Products	☒ ET Tubes	☒ Other: intralipid	
☐ Intraosseous Set-up	☐ LMA	☐ Other:	
D. Moulage			
None required. Could put emesis on shirt if desired.			
E. Approximate Timing			
Set-Up:	3 min	Scenario:	12 min
		Debriefing:	15 min



Section V: Patient Data and Baseline State

A. Clinical Vignette: To Read Aloud at Beginning of Case

A 48-year-old female presents to the ED with an unknown overdose. She was out drinking with friends until an hour ago. Her daughter came home and found her with vomit around her, empty pill bottles, and bits of pills in her vomit.

B. Patient Profile and History

Patient Name: Yvette Clark	Age: 48	Weight: 100kg
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Gender: ☐ M ☒ F	Code Status: Full
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Chief Complaint: ?OD

History of Presenting Illness: Out drinking last night. May have overdosed on meds when got home.

Past Medical History:	HTN	Medications:	Nifedipine XR 60mg daily
	DM		Glyburide 5mg daily
	Depression		Lasix 40mg daily
			HCTZ 25mg daily
			Clonazepam 2mg qHS

Allergies: None

Social History: Single mother of two children, now grown. (Daughter here in ED.) Drinks heavily on weekends.

Review of Systems:	CNS:	Nil		
	HEENT:	Nil		
	CVS:	Nil.		
	RESP:	Nil		
	GI:	Emesis prior to arrival		
	GU:	Nil		
	MSK:	Nil	INT:	Nil

C. Baseline Simulator State and Physical Exam

☐ No Monitor Display	☒ Monitor On, no data displayed	☐ Monitor on Standard Display
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HR: 50/min	BP: 87/43	RR: 10/min	O ₂ SAT: 95% RA
Rhythm: junctional	T: 36°C	Glucose: 6.8 mmol/L	GCS: 9 (E2 V2 M5)

General Status: Somnolent, looks unwell.

CNS:	Somnolent. Grunts/moans only. Opens eyes to pain, localizes to pain.		
HEENT:	No signs HI. Pupils 3mm, reactive.		
CVS:	No murmur.		
RESP:	Slow resp rate. Slight work of breathing.		
ABDO:	Soft, NT.		
GU:	Nil.		
MSK:	No signs trauma.	SKIN:	Nil.



Section VI: Scenario Progression

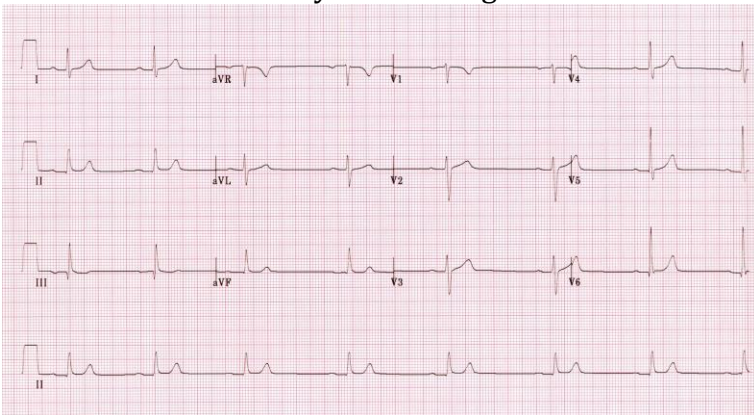
Scenario States, Modifiers and Triggers			
Patient State	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State	
1. Baseline State Rhythm: sinus brady with 1 st degree block HR: 50/min BP: 87/43 RR: 10/min O ₂ SAT: 95% RA T: 36°C	Somnolent, responding to pain only.	<u>Learner Actions</u> ☐ IV, O ₂ , monitors ☐ IV NS 1L bolus ☐ Tox bloodwork, VBG ☐ ECG ☐ Go through med list ☐ More history/px exam ☐ Check glucose (6.8) ☐ Trial narcans ☐ Call poison centre	<u>Modifiers</u> <i>Changes to patient condition based on learner action</i> - Narcan → no effect - 1L bolus → BP 92/50 transiently - Glucagon given → emesis <u>Triggers</u> <i>For progression to next state</i> - 3 min → 2. Persistent Hypotension
2. Persistent Hypotension BP → 75/45	Patient status unchanged	<u>Learner Actions</u> ☐ Ca gluconate 2g iv bolus, consider Ca infusion 2-3g/hr ☐ Insulin infusion (1u/kg/hr with 0.5u/kg bolus) with D10NS @ 150ml/hr ☐ Check glucose q15 min ☐ Repeat NS bolus ☐ Add vasopressor infusion	<u>Modifiers</u> - Post calcium → BP 83/46 <u>Triggers</u> - All 3 agents started → 3. More Somnolent - 9 min → 3. More Somnolent
3. More Somnolent RR → 6 O ₂ SAT → 91%	Begin state with RN saying "she seems to be making less respiratory effort"	<u>Learner Actions</u> ☐ Check glucose (7.1) ☐ Trial narcans (if not before) ☐ Assist breathing (bagging) ☐ Prepare for intubation	<u>Modifiers</u> - Narcan → No effect <u>Triggers</u> - Intubation → 4. Intubation - If don't intubate → give VBG. At 12 min → 5. Arrest
4. Intubation With pre-oxygenation: O ₂ SAT → 95% Post paralytic: RR → 0		<u>Learner Actions</u> ☐ Appropriate RSI ☐ Post-intubation CXR ☐ Sedation infusion ☐ Prepare for central line ☐ OG placement ☐ Call ICU	<u>Modifiers</u> - Propofol for intubation → BP 50/22 <u>Triggers</u> - Intubated → 5. Arrest
5. Arrest Rhythm: asystole No vitals	Patient pulseless.	<u>Learner Actions</u> ☐ High quality CPR ☐ EtCO ₂ monitoring ☐ Epinephrine q3 min ☐ Calcium Chloride & HCO ₃ ☐ Consider intralipid	<u>Modifiers</u> <u>Triggers</u> - Intralipid given → END CASE - 15 min → END CASE



Section VII: Supporting Documents, Laboratory Results, & Multimedia

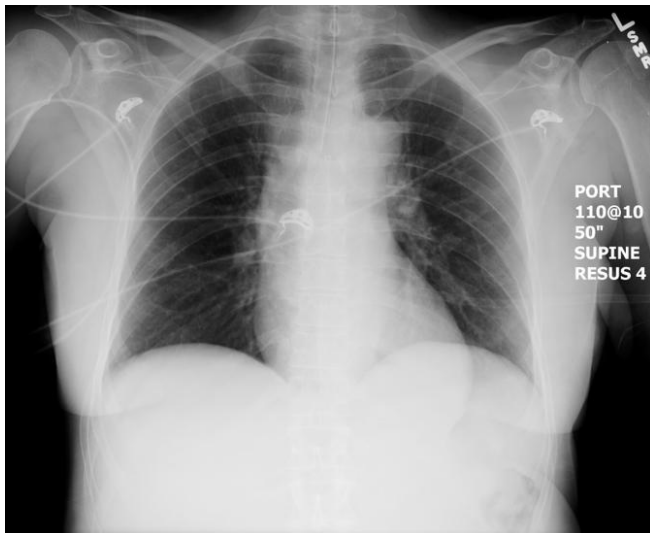
Laboratory Results					
VBG	pH: 6.97	PCO ₂ : 106	PO ₂ : 50	HCO ₃ : 7	

Images (ECGs, CXRs, etc.)

Initial ECG: Sinus brady with 1st degree block

ECG source: <http://lifeinthefastlane.com/ecg-library/beta-blocker-and-calcium-channel-blocker-toxicity/sb-1hb/>

Post-Intubation CXR:



CXR source: <https://emcow.files.wordpress.com/2012/11/normal-intubation2.jpg>



Section VIII: Debriefing Guide

General Debriefing Plan	
☐ Individual	☒ Group
☐ With Video	☒ Without Video
Objectives	
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Sample Questions for Debriefing	
1) When did you first consider nifedipine to be a possible cause of hypotension? 2) When did you first consider intubation? 3) What are the first line treatments for a CCB overdose? 4) When is intralipid indicated in a CCB overdose? 5) Why do you think the patient became somnolent?	
Key Moments	
Recognition of hypotension as possibly due to CCB	
Recognition of respiratory depression and need for intubation	
Recognition of need for intralipid in context of cardiac arrest	